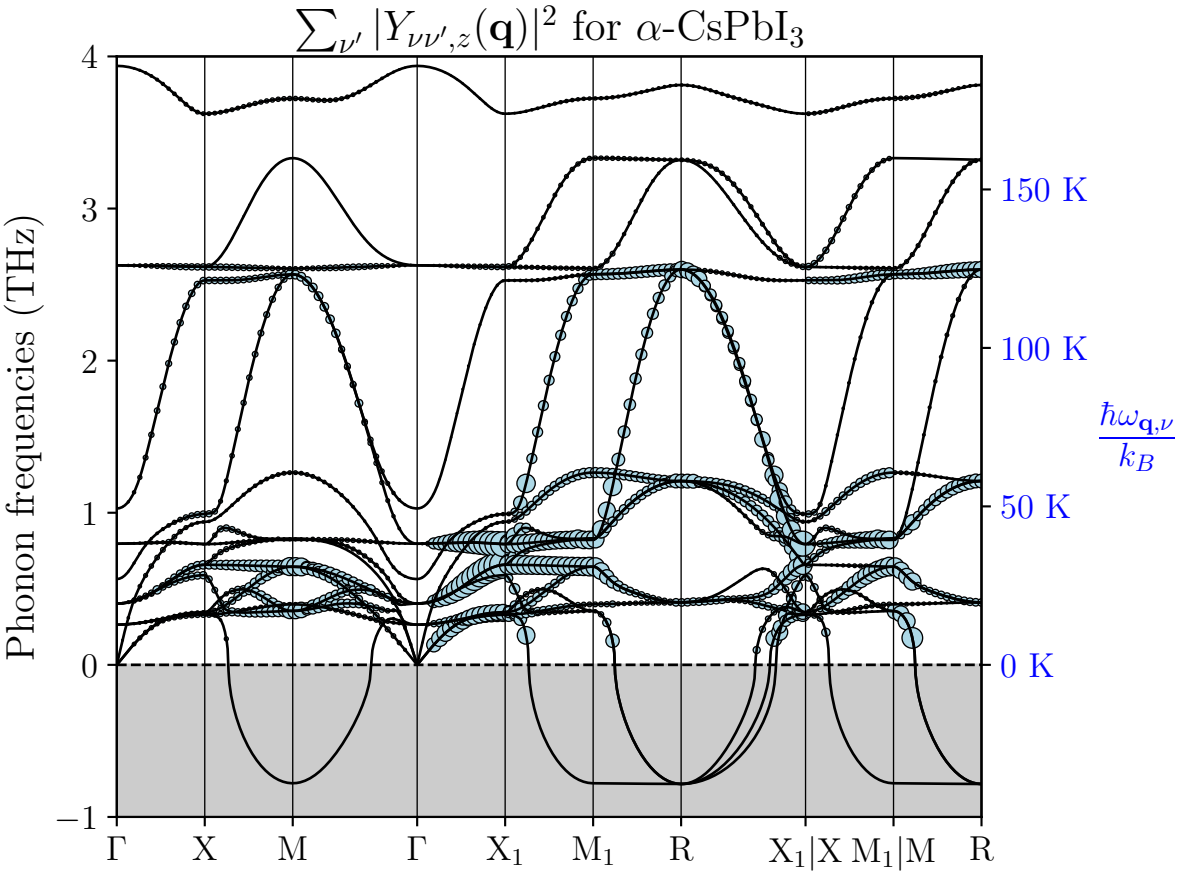
$\sum_{\nu'} |Y_{\nu\nu',z}(\mathbf{q})|^2$ for α-$CsPbI_3$
Phonon frequencies (THz)
4
3
2
1
0
−1
150 K
100 K
50 K
0 K
$\frac{\hbar\omega_{\mathbf{q},\nu}}{k_B}$
Γ
X
M
Γ
X_1
M_1
R
X_1|X
M_1|M
R